

Zhao Zehui, Leila

🌐 <https://github.com/Leilazehui/Leilazehui.github.io>

☎ +85297130803

✉ leilazhao853@gmail.com

🌐 <https://www.linkedin.com/in/zehui-zhao>

EDUCATION

The University of Hong Kong, Bachelor's in Mechanical Engineering

2022 – now

- Highest Attained GPA: 3.36/4.3
- Scholarship: Tam Wing Fan Innovation Academy Funding 2025 – 2026
- Course taken: Mechanics of Solid (A), Engineering design and manufacture (A+), Unmanned Aerial Vehicle design (A+), Dynamics and Control (A-), Thermodynamics (A-), Advanced Partial Differential Equation (B-), Mechanics of Fluid (B-), Electrical Information and Quantum Physics (B-)

WORK EXPERIENCES

The Hong Kong Productivity Council

Hong Kong

Winter Intern in Smart Manufacturing Division

Dec 2025 - now

- Researched global textile recycling regulations and standards and developed an ESG toolkit to drive circular innovation and long-term sustainability for Hong Kong fashion enterprises, which provide valuable data base for Hong Kong SMEs in green recycling and integration of AI field, and prepared survey to collect market information about AI application in footwear industry's operation.
- Designed a 3D camera plus multi-axis robotic arm mounting system, applying 2D, 3D drawing, BOM and GD&T standard for manufacturing via SolidWorks.
- Conducting a thermographic camera project to classify normal and abnormal temperature region on the robotic arm during its operation by training an AI model on Roboflow.
- Designed an indoor drone inspection project for detecting inner cracks of building using SLAM navigation, visual inspection and extracting mechanical data from FEA simulation for AI application

Faculty of Engineering, the University of Hong Kong

Hong Kong

Part-time Temporary Student Research Assistant

Sept – Nov 2025

- Mentored over 30 students in robotic arm control and design, enhancing their practical skills and understanding of engineering concepts.
- Provide constructive feedback to lecturers on behalf of workshop content and delivery which assisted them in workshop development.

Feelings Group Limited | Hong Kong Science & Technology Park

Hong Kong

Student Technician – Hardware team

Dec 2024 – Jan 2025

- Conducted literature study and market research for Radar hardware systems medical device development for tracking vital signs of small animals, provide academic and industrial research, market database for the project, which is now manufactured and promoted in exhibitions in Dubai, US and other countries
- Assisted in camera development using electronic modules, such as Raspberry Pi, LAN network and camera module for CCTV development for the medical device.

The Hong Kong and China Gas Company Limited (Towngas)

Hong Kong

Summer intern in residential product & development department

Jul - Aug 2024

- Contributed to the development of 7 IoT gas appliances, and more than 10 gateways.
- Identified potential problems in functional tests and provided effective suggestions to enhance user-friendliness of the mobile app interface.
- Assisted in the assessment of gas appliance accessories such as flexible hose and gas malleable iron fittings.
- Analyzed market potential for company's products and participated in daily meetings, contributed to strategic planning and understanding team structure and working principles and provided valuable feedback to supervisor regarding the market demand and daily workflow.

PROJECTS & COMPETITIONS

HKU Design, Build and Fly Team

Hong Kong, USA

HKU AIAA DBF Competition Team Leader

Aug 2025 – now

- Optimized main wing specification under the constraint of 1.5-meter wingspan, to provide sufficient lift for an 8kg aircraft and fast-speed flight.
- Conducted Aerodynamic analysis on airfoils and wings on XF1r5 to verify the stability of the aircraft and ease the manufacturing process.
- Performed CFD analysis on Ansys on entire model aircraft and provided valuable feedback for overall aircraft design, particularly banner mechanism and empennage section.
- Designed banner towing mechanisms by utilizing CAD, electronics and 3D printing for prototypes, deployed 2-m long banner in flight.
- Run and examine an aircraft Performance analysis by automating the calculation process for mission score optimization and computed the best score case for the team aircraft configuration on MATLAB.

HKU DBF Student Interest Group Leader

Sept 2024 – now

- Conducted over 30 workshops and training to promote the team to new members, and received 100K HKD funding from Innovation Academy, HKU.

- Secured three industry sponsorships, including Ansys, L'Voyage and the Safety Collaborative for DBF2025-2026.
- Manage the team's human resources, documents, promotion and train next year's leaders to enlarge the team size and development.
- Conducted research study on 3D printing material regarding the topic of lightweight design and analysis for ribs construction in fixed-wing model aircraft, tested around 5-8 materials and provide the most ideal solution for lightweight fixed-wing design.

Cathay Pacific Career Experience Day

Oct 2025

- A 1-hour mock hackathon in CX city which I worked with 3 other students to come up with an 800K project proposal; oversaw product design which aims for providing scented pillows as mental comfort and relaxation for travelers and conducted market research and financial planning, received the second highest feedback from judges.

MATLB Interstellar Project

Dec 2024

- Analyze stellar motion by data plotting regarding intensity and wavelength of 7 stars to calculate their doppler effect, and compare their stellar spectra provided in the database.

RAICOM Robotic Arm Challenge (Shenzhen)

Jul 2025

- Broke down the structure of the algorithm for automated and semi-automated missions; performed data annotation for AI model training.

Hong Kong Techathon

Sept 2023

- Provided project ideas and designed an app interface for educational purposes in subjects related to Astronomy.

CERTIFICATES & EXTRA-CURRICULAR ACTIVITIES

Model-Based System Engineering Fundamentals, Coursera, University at Buffalo, State University of New York 2025

Royal Aeronautical Society (HK) Student Member 2025

IMechE Affiliated Member & Committee 2025

General Secretary, Publication and Publicity Secretary, Physics Society, HKU (2023)

- Managed the documents, meeting minutes for annual general meetings, promotions on social media and designed posts using Canva.
- Secured 3000 HKD fundings from professors in the Physics Department for secondary-school promotion events support.

HKIE Aircraft Division – Student Member

Peking University Summer Exchange Program 2023

Junior Student Ambassador of Equal Opportunity Unit, HKU 2023

HKU Lead-For-Life Cohort Member 2022 – 2026

Grade 7, IELTS EXAM 2023

Champion in Public Group, The 1st “National Aerospace Dream”, Hong Kong Aerospace Knowledge Competition 2022

Member of Astronomical Training Program for Secondary Student, 2019 - 2020

SKILLS

Language: Chinese (Mandarin, Cantonese): Native, English: Proficient, German: Basic

CAD Software:

SolidWorks (Proficient), AutoCAD (Proficient), fusion360 (Working level)

CFD Software:

Xflr5 (Proficient), Ansys (Working level)

Other Software:

HIKIMICRO Studio (Working level), Bambu lab Studio (intermediate), Microsoft Office Suite (Intermediate), Roboflow

Programming:

MATLAB (Working level), Arduino (Working level), Python (Working level), Orange data mining (Working level), Visual studio (working level)

INTERESTS & VOLUNTEERING

Music: Member and the Head of Sanxian Department, Hong Kong Youth Chinese Orchestra, 2016 – 2023; Zheng Grading Class Diploma – Champion - HKYPA, 2020; Performance Diploma Guzheng Exam – distinction - The Central Conservatory of Music, 2018; Joint School Music Competition – Symphony Orchestra (Secondary) - Gold Award, 2018;

Business: Member of JA Company – 2nd Runner-up, 2019

Volunteering: Volunteering Performers in HKPolyU 2023; Performer of postcard design competition ceremony by HK education Bureau, 2019; performer in Mirrorpost traditional culture promotion event, 2016